

Deep Learning-based mmWave MIMO Channel Estimation using sub-6 GHz Channel Information: CNN and UNet Approaches

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Abstract—Future wireless multiple-input multiple-output (MIMO) systems will integrate both sub-6 GHz and millimeter wave (mmWave) frequency bands to meet the growing demands for high data rates. MIMO link establishment typically requires accurate channel estimation, which is particularly challenging at mmWave frequencies due to the low signal-to-noise ratio (SNR). In this paper, we propose two novel deep learning-based methods for estimating mmWave MIMO channels by leveraging out-of-band information from the sub-6 GHz band. The first method employs a convolutional neural network (CNN), while the second method utilizes a UNet architecture. We compare these proposed methods against deep-learning methods that rely solely on in-band information and with other state-of-the-art out-of-band aided methods. Simulation results show that our proposed out-of-band aided deep-learning methods outperform existing alternatives in terms of achievable spectral efficiency.

Index Terms—mmWave, MIMO, channel estimation, deep learning, CNN, UNet.

I. INTRODUCTION

Millimeter wave (mmWave) communications have the potential to meet high data rate demands due to their extensive bandwidth [1], [2]. To overcome high propagation losses and ensure sufficient link margin, most mmWave systems will employ large antenna arrays and multiple-input multiple-output (MIMO) techniques [3]. The key challenge in implementing mmWave MIMO systems is link establishment, which relies on accurate channel estimation [3]. However, at mmWave frequencies, channel estimation is particularly challenging due to the low pre-beamforming signal-to-noise ratio (SNR).

Recently, machine learning (ML) techniques based on the multi-layer perceptrons, commonly referred as deep learning, have attracted significant attention for mmWave MIMO channel estimation [4]–[6]. Authors have utilized deep convolutional neural network (CNN) [4], fast and flexible denoising CNN (FFDNet) [5], and ResNet-UNet structure [6] to address channel estimation in mmWave MIMO systems. However, these methods rely solely on in-band mmWave chan-

nel information, which is inherently constrained by low pre-beamforming SNR, thereby limiting their overall performance.

Furthermore, mmWave systems are being implemented in conjunction with sub-6 GHz systems to facilitate multi-band communication [7]. Multiple multi-band measurement campaigns in various environments have revealed that sub-6 GHz and mmWave frequency bands share similar multipath characteristics [8]–[11]. However, systems operating below 6 GHz exhibit lower propagation loss, resulting in a higher pre-beamforming SNR. This reliable out-of-band information from sub-6 GHz can be utilized to support and enhance channel estimation for mmWave systems. The use of out-of-band information for mmWave channel estimation has been investigated in [12], [13]. These studies propose novel pilot-aided channel estimation methods for mmWave MIMO systems, utilizing the line-of-sight (LOS) channel component acquired with the support of the sub-6 GHz band. However, without incorporating ML, many inherent relationships between in-band and out-of-band channel estimates remain unexplored.

Contribution: In this paper, we improve the out-of-band aided channel estimation methods from [13] by incorporating ML. Unlike conventional methods, ML has a greater capacity to uncover the intrinsic relationships between in-band and out-of-band channel estimates. We propose two novel methods for mmWave MIMO systems: one based on a CNN and the other employing a UNet architecture. We evaluate the proposed methods through simulations in terms of normalized channel mean squared error (MSE) and spectral efficiency (SE).

Organization: Section II presents the system model. Section III outlines the channel estimation methods, followed by their simulation-based evaluation in Section IV. Finally, Section V concludes the paper.

Notation: The superscript $(\cdot)^{(b)}$ represents frequency-band dependent values, where $b \in \{s, m\}$, with s referring to the sub-6 GHz and m to the mmWave band. Scalars are denoted by x , vectors by bold lowercase letters $\mathbf{x}$ and matrices by bold uppercase letters $\mathbf{X}$. The i -th row of the matrix $\mathbf{X}$ is indicated by $\mathbf{X}_{i,:}$, and the j -th column by $\mathbf{X}_{:,j}$. The all-ones matrix is denoted by $\mathbf{1}$, with its dimensions specified in the subscript. The superscripts $(\cdot)^T$ and $(\cdot)^H$ represent transpose and Hermitian transpose, respectively. The Euclidean norm is represented by $\|\cdot\|$ and the Frobenius norm by $\|\cdot\|_F$.

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II. SYSTEM MODEL

We consider a point-to-point MIMO system that operates simultaneously on both sub-6 GHz and mmWave frequency bands in the radiative far-field regime. Both the sub-6 GHz and mmWave arrays are arranged in a uniform linear array (ULA) configuration, co-located and aligned to ensure identical angle-of-departure (AoD) ϑ and angle-of-arrival (AoA) φ for both arrays (see Fig. 1). We assume perfect synchronization between the transmitters and receivers in both time and frequency domains. Each transmitter and receiver is equipped with a single radio frequency (RF) chain per antenna, enabling fully digital beamforming at both the sub-6 GHz and mmWave frequency bands. The system operates using a time-division duplex (TDD) protocol, ensuring reciprocal channel responses. The transmission system employs an orthogonal frequency-division multiplexing (OFDM) system and quadrature amplitude modulation with $N^{(b)}$ subcarriers.

A. Channel Model

At the n -th OFDM subcarrier, the channel is represented by the matrix $\mathbf{H}^{(b)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(b)} \times M_{\text{Tx}}^{(b)}}$, based on the equivalent OFDM complex baseband representation. The channel follows a frequency-selective Rician fading channel model [14]

$$\mathbf{H}^{(b)}[n] = \sqrt{\eta^{(b)}} \sqrt{\frac{K^{(b)}}{1 + K^{(b)}}} \underbrace{\mathbf{H}_{\text{fs}}^{(b)}}_{\text{free-space}} + \sqrt{\eta^{(b)}} \sqrt{\frac{1}{1 + K^{(b)}}} \underbrace{\mathbf{H}_{\text{rp}}^{(b)}[n]}_{\text{Rayleigh part}}, \quad (1)$$

where $\eta^{(b)}$ represents the path gain including shadowing and $K^{(b)}$ indicates the Rician K -factor for the band b . Given the different multipath characteristics between mmWave and sub-6 GHz bands, we assume that the K -factors for these bands are related by $K^{(m)} = c_K K^{(s)}$, where c_K is the scaling factor.

The deterministic LOS channel $\mathbf{H}_{\text{fs}}^{(b)}[n]$ is given by

$$\mathbf{H}_{\text{fs}}^{(b)} = e^{j\chi^{(b)}} \mathbf{a}_{\text{Rx}}^{(b)}(\varphi) \left(\mathbf{a}_{\text{Tx}}^{(b)}(\vartheta) \right)^H, \quad (2)$$

where

$$\mathbf{a}_{\text{Tx}}^{(b)}(\vartheta) = \begin{bmatrix} 1 & \dots & e^{-j2\pi(M_{\text{Tx}}^{(b)}-1)\frac{\Delta d^{(b)}}{\lambda^{(b)}}\sin(\vartheta)} \end{bmatrix}^T \quad (3)$$

and

$$\mathbf{a}_{\text{Rx}}^{(b)}(\varphi) = \begin{bmatrix} 1 & \dots & e^{-j2\pi(M_{\text{Rx}}^{(b)}-1)\frac{\Delta d^{(b)}}{\lambda^{(b)}}\sin(\varphi)} \end{bmatrix}^T \quad (4)$$

are the steering vectors corresponding to the AoD ϑ and the AoA φ , respectively. In (2), $\chi^{(b)} \sim \mathcal{U}(-\pi, \pi)$ denotes the initial random phase offset of the free-space LOS channel.

The Rayleigh matrix $\mathbf{H}_{\text{rp}}^{(b)}[n]$ is modeled following the 3rd Generation Partnership Project (3GPP) MIMO channel model [15]. Due to the significant differences between sub-6 GHz and mmWave wavelengths, the diffuse components $\mathbf{H}_{\text{rp}}^{(s)}[n]$ and $\mathbf{H}_{\text{rp}}^{(m)}[n]$ are assumed to be completely uncorrelated. The channel matrix $\mathbf{H}^{(b)}[\tau]$ in the delay domain for the delay-tap τ is then derived by applying the inverse discrete Fourier transform (IDFT) to $\mathbf{H}^{(b)}[n]$.

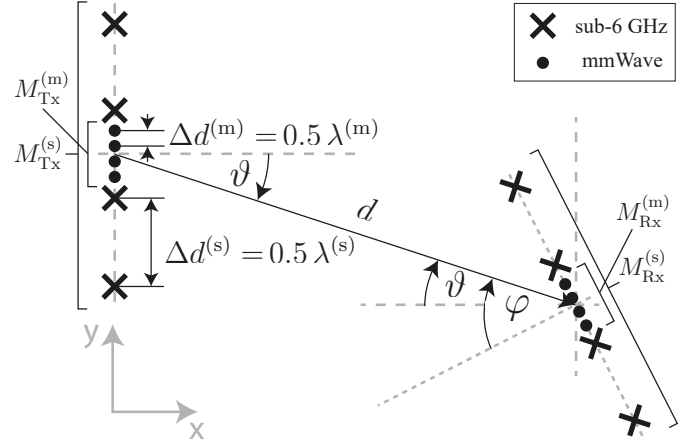


Fig. 1. A point-to-point MIMO system with co-located sub-6 GHz and mmWave antenna arrays. The system geometry is characterized by the number of transmit antennas $M_{\text{Tx}}^{(b)}$, receive antennas $M_{\text{Rx}}^{(b)}$ and their mutual separation $\Delta d^{(b)}$, along with the corresponding AoD ϑ and AoA φ .

B. Link Establishment

Link establishment involves a training phase followed by data transmission. In the training phase, we first estimate the wireless channel at both the sub-6 GHz and mmWave frequency bands. These channel estimates are then combined and utilized for data transmission over the mmWave band.

1) *Training Phase:* The training phase consists of two sequential steps. In the first step, we transmit non-precoded pilot symbols across both frequency bands. The input-output relationship for this training step is expressed as

$$\mathbf{y}^{(b)}[n] = \mathbf{H}^{(b)}[n]\phi^{(b)}[n] + \mathbf{w}^{(b)}[n], \quad (5)$$

where $\mathbf{y}^{(b)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(b)} \times 1}$ denotes the received symbols and $\mathbf{w}^{(b)}[n] \in \mathcal{CN}(0, \sigma_{w^{(b)}}^2 \mathbf{I}_{M_{\text{Rx}}^{(b)}})$ represents the additive white Gaussian noise (AWGN). In (5), the pilot symbols for the t -th transmit antenna are defined as follows:

$$\phi_t^{(b)}[n] = \begin{cases} \phi^{(b)}[n], & n \in \{t, t + M_{\text{Tx}}^{(b)}, \dots, N^{(b)}\} \\ 0, & \text{else} \end{cases} \quad (6)$$

At the receiver, the channel estimate $\tilde{\mathbf{H}}^{(b)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(b)} \times M_{\text{Tx}}^{(b)}}$ is obtained using least-squares (LS) estimation, followed by linear interpolation. To enable optimal beamforming matrix selection, the channel estimated at the receiver must also be available at the transmitter. Given that the system operates using the TDD protocol, channel reciprocity in both frequency bands is utilized to enable acquisition of the channel estimate at the transmitter.

In the second step, the sub-6 GHz channel estimate $\tilde{\mathbf{H}}^{(s)}[n]$ is used to jointly estimate the AoA and AoD of the LOS component using the estimation method described in [13]. Using this estimated angular information $\tilde{\varphi}$ and $\tilde{\vartheta}$, we transmit precoded/combined pilot symbols in the direction of the LOS

component within the mmWave band only. The input-output relationship for this step is given by

$$\mathbf{y}^{(m)}[n] = \underbrace{\left(\mathbf{a}_{\text{Rx}}^{(m)}(\tilde{\varphi})\right)^H \mathbf{H}^{(m)}[n] \mathbf{a}_{\text{Tx}}^{(m)}(\tilde{\vartheta})}_{\mathbf{G}^{(m)}[n]} \phi^{(m)}[n] + \left(\mathbf{a}_{\text{Rx}}^{(m)}(\tilde{\varphi})\right)^H \mathbf{w}^{(m)}[n], \quad (7)$$

where $\phi^{(m)}[n]$ and $\mathbf{y}^{(m)}[n]$ are one-dimensional complex-valued variables, given that only a single spatial stream is used. Next, LS estimation is applied, followed by linear interpolation, to obtain the effective beamformed scalar channel estimate $\hat{\mathbf{G}}^{(m)}[n]$. The LOS component of $\hat{\mathbf{G}}^{(m)}[n]$ is then isolated through delay-domain filtering $\hat{\mathbf{G}}_{\text{LOS}}^{(m)}[\tau] = \hat{\mathbf{G}}^{(m)}[\tau]\delta[\tau]$, where $\delta[\cdot]$ denotes the delta function. Finally, the LOS channel estimate $\hat{\mathbf{G}}_{\text{LOS}}^{(m)}[\tau]$ is transformed back to the frequency domain via the discrete Fourier transform (DFT). The resulting out-of-band aided channel estimate for the LOS component is reconstructed as

$$\hat{\mathbf{H}}^{(m)}[n] = \mathbf{a}_{\text{Rx}}^{(m)}(\tilde{\varphi}) \hat{\mathbf{G}}_{\text{LOS}}^{(m)}[n] \left(\mathbf{a}_{\text{Tx}}^{(m)}(\tilde{\vartheta})\right)^H. \quad (8)$$

2) *Data Transmission*: We first process the channel estimates obtained during the training phase using the methods described in Section III. The resulting mmWave channel estimate $\bar{\mathbf{H}}^{(m)}[n]$ is then used for precoding and combining. To achieve optimal performance in terms of SE, we apply singular value decomposition (SVD) to $\bar{\mathbf{H}}^{(m)}[n]$, with its compact form given by

$$\bar{\mathbf{H}}^{(m)}[n] = \bar{\mathbf{Q}}^{(m)}[n] \bar{\mathbf{\Sigma}}^{(m)}[n] \left(\bar{\mathbf{F}}^{(m)}[n]\right)^H. \quad (9)$$

In (9), $\bar{\mathbf{Q}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(m)} \times \ell_{\text{max}}}$ is the matrix of left singular vectors, $\bar{\mathbf{F}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Tx}}^{(m)} \times \ell_{\text{max}}}$ is the matrix of right singular vectors, $\ell_{\text{max}} = \min(M_{\text{Rx}}^{(m)}, M_{\text{Tx}}^{(m)})$ is the maximum number of streams and $\bar{\mathbf{\Sigma}}^{(m)}[n]$ is the diagonal matrix of singular values $\bar{\sigma}_{(1)}^{(m)}[n], \dots, \bar{\sigma}_{(\ell_{\text{max}})}^{(m)}[n]$. The power loading matrix, characterized by diagonal elements $\bar{p}_{(1)}^{(m)}[n], \dots, \bar{p}_{(\ell_{\text{max}})}^{(m)}[n]$, is optimized to maximize the transmission rate by following the water-filling power allocation policy [16]. Further, the precoding matrix ensures compliance with the total transmit power constraint

$$\left\| \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n]\right)^{1/2} \right\|_F^2 = P_{\text{T}}^{(m)}. \quad (10)$$

The input-output relationship for the data transmission phase is given by

$$\mathbf{y}^{(m)}[n] = \left(\bar{\mathbf{Q}}^{(m)}[n]\right)^H \mathbf{H}^{(m)}[n] \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n]\right)^{1/2} \mathbf{x}^{(m)}[n] + \left(\bar{\mathbf{Q}}^{(m)}[n]\right)^H \mathbf{w}^{(m)}[n], \quad (11)$$

where $\mathbf{x}^{(m)}[n] \in \mathbb{C}^{\ell_{\text{max}} \times 1}$ denotes the transmit symbol vector, $\mathbf{y}^{(m)}[n] \in \mathbb{C}^{\ell_{\text{max}} \times 1}$ represents the received symbol vector and $\mathbf{w}^{(m)}[n] \in \mathcal{CN}(0, \sigma_{w^{(m)}}^2 \mathbf{I}_{M_{\text{Rx}}^{(m)}})$ is the AWGN.

III. CHANNEL ESTIMATION METHODS

In this section, we introduce channel estimation methods for the mmWave system, combining in-band $\hat{\mathbf{H}}^{(m)}[n]$ and out-of-band aided $\hat{\mathbf{H}}^{(m)}[n]$ channel estimates to improve estimation quality. We first revisit the maximal ratio combining (MRC) method, which is the best-performing method from [13] in terms of SE. Following this, we propose two novel channel estimation methods that combine the available estimates using ML algorithms.

A. Maximal Ratio Combining (MRC)

This method combines the out-of-band aided channel estimate $\hat{\mathbf{H}}^{(m)}[n]$ and the in-band estimate $\hat{\mathbf{H}}^{(m)}[n]$ using MRC, given by

$$\bar{\mathbf{H}}^{(m)}[n] = \hat{w} \hat{\mathbf{H}}^{(m)}[n] + (1 - \hat{w}) \tilde{\mathbf{H}}^{(m)}[n], \quad (12)$$

where

$$\hat{w} \approx \frac{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)} \sigma_{w^{(m)}}^2}{\frac{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}}{1 + c_K \tilde{K}^{(s)}} + \left(1 + M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}\right) \sigma_{w^{(m)}}^2}. \quad (13)$$

denotes the approximated optimal combining factor, with its detailed derivation provided in [13]. In (13), $\tilde{K}^{(s)}$ represents the K -factor for the sub-6 GHz band, estimated using the method of moments [17]. This approach is designed to adapt to varying wireless channel conditions, particularly in relation to the K -factor and the noise variance $\sigma_{w^{(m)}}^2$.

B. Deep Learning-based Methods

With the aim of improving the performance of the proposed MRC method in terms of SE, we apply CNN-based and UNet-based channel estimation, exploiting in-band and out-of-band aided channel estimates. The deployment of both the CNN and UNet consists of two phases: the offline training and the online deployment.

1) *CNN-based Method*: CNNs are particularly well suited for MIMO systems, where the spatial arrangement of antennas plays a crucial role [4]. As input to the CNN, we use the in-band $\hat{\mathbf{H}}^{(m)}[n]$ and out-of-band aided $\tilde{\mathbf{H}}^{(m)}[n]$ channel estimates. Furthermore, as in (13), we provide as input the estimated sub-6 GHz K -factor matrix $\tilde{\mathbf{K}}^{(s)} = \tilde{K}^{(s)} \mathbf{1}_{M_{\text{Rx}}^{(m)} \times M_{\text{Tx}}^{(m)}}$ and the noise variance matrix $\mathbf{\Gamma}^{(m)} = \sigma_{w^{(m)}}^2 \mathbf{1}_{M_{\text{Rx}}^{(m)} \times M_{\text{Tx}}^{(m)}}$. We assume that the noise variance $\sigma_{w^{(m)}}^2$ is known at the receiver and that is related to pre-beamforming SNR with $\gamma^{(m)} = 1/\sigma_{w^{(m)}}^2$. The CNN outputs the estimated channel matrix $\bar{\mathbf{H}}^{(m)}[n]$ through the mapping relationship

$$\bar{\mathbf{H}}^{(m)}[n] = f_{\Phi_C} \left(\tilde{\mathbf{H}}^{(m)}[n], \hat{\mathbf{H}}^{(m)}[n], \tilde{\mathbf{K}}^{(s)}, \mathbf{\Gamma}^{(m)}; \Phi_C \right), \quad (14)$$

where Φ_C denotes the parameter set of the CNN. The block diagram of the CNN architecture is illustrated in Fig. 2. Each intermediate layer employs the rectified linear unit (ReLU) activation function, followed by a batch normalization (BN) layer to stabilize and accelerate the learning process. Additionally, zero padding (ZP) is applied to maintain the dimensions of the feature matrix during convolution. The output layer

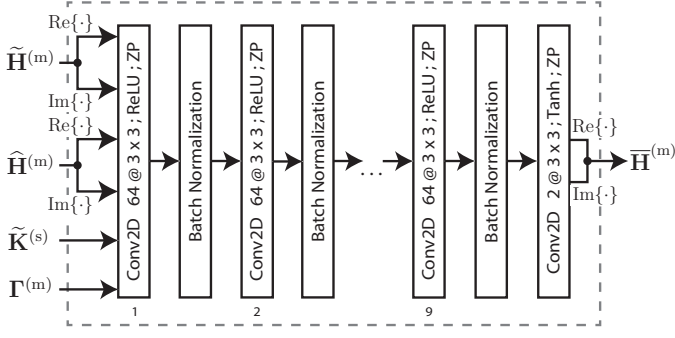


Fig. 2. The CNN architecture comprises nine convolutional layers, followed by an output layer that generates the estimated channel matrix $\bar{\mathbf{H}}^{(m)}[n]$.

applies a hyperbolic tangent activation function to generate the estimated real and imaginary parts of the channel matrices.

2) *UNet-based Method*: To further enhance performance in terms of SE, we employ UNet-based channel estimation. UNet offers advantages over standard CNNs, due to its ability to capture both global information and fine-grained details via skip connections and its encoder-decoder structure. In [6], UNet demonstrated greater robustness to noise compared to standard CNNs. The UNet takes the same input as the CNN and outputs the estimated channel matrix $\bar{\mathbf{H}}^{(m)}[n]$ through the mapping relationship

$$\bar{\mathbf{H}}^{(m)}[n] = f_{\Phi_U}(\tilde{\mathbf{H}}^{(m)}[n], \hat{\mathbf{H}}^{(m)}[n], \tilde{\mathbf{K}}^{(s)}, \mathbf{\Gamma}^{(m)}; \Phi_U), \quad (15)$$

where Φ_U denotes the parameter set of the UNet. The block diagram of the UNet architecture is shown in Fig. 3. Each encoder and decoder comprises five convolutional layers, each followed by a BN layer. The contraction path captures contextual information, while the symmetric expansion path enables precise spatial localization [6]. Skip connections are employed to directly link encoder layers to corresponding decoder layers, enabling the transfer of features extracted during encoding to enhance decoding performance. At the end of the expansion path, the final decoder block is used to generate the output. The output layer applies a hyperbolic tangent activation function to generate the estimated real and imaginary parts of the channel matrices.

3) *Offline Training*: For the proposed networks, the training set consisting of L_{train} samples is generated according to a specific channel model in the simulation environment. The l -th sample is represented as $(\tilde{\mathbf{H}}_l^{(m)}, \hat{\mathbf{H}}_l^{(m)}, \tilde{\mathbf{K}}_l^{(s)}, \mathbf{\Gamma}_l^{(m)}; \mathbf{H}_l^{(m)})$, where $\tilde{\mathbf{H}}_l^{(m)}, \hat{\mathbf{H}}_l^{(m)}, \tilde{\mathbf{K}}_l^{(s)}$ and $\mathbf{\Gamma}_l^{(m)}$ are input data, and $\mathbf{H}_l^{(m)}$ is the target data. To reduce the complexity of the model, we consider only the single central subcarrier ($n = N^{(m)}/2$) for both the input and target data. Before entering the network, the $\tilde{\mathbf{H}}_l^{(m)}$ and $\hat{\mathbf{H}}_l^{(m)}$ are separated into their real and imaginary parts. As input, the CNN and UNet take six real-valued matrices of dimension $M_{\text{Rx}}^{(m)} \times M_{\text{Tx}}^{(m)}$. At the output, the real and imaginary parts are recombined to obtain the $M_{\text{Rx}}^{(m)} \times M_{\text{Tx}}^{(m)}$ complex-valued estimated channel matrix $\bar{\mathbf{H}}_l^{(m)}$. The goal of

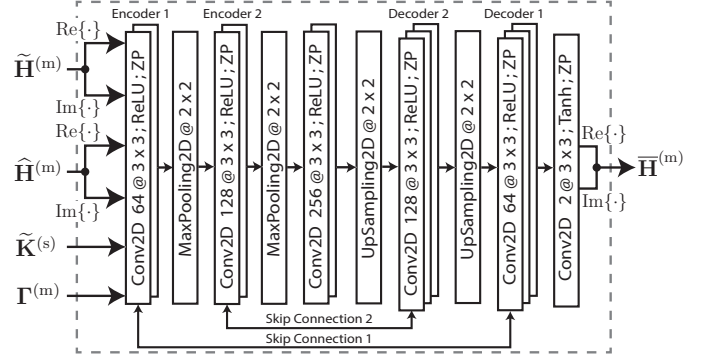


Fig. 3. The UNet architecture consists of two encoders in the contraction path and two decoders in the symmetric expansion path, followed by an output layer that produces the estimated channel matrix $\bar{\mathbf{H}}^{(m)}[n]$.

the offline training is to minimize the normalized channel estimation MSE loss function

$$\text{NMSE}_{\text{Loss}} = \frac{1}{L_{\text{train}}} \sum_{l=1}^{L_{\text{train}}} \frac{\|\mathbf{H}_l^{(m)} - \bar{\mathbf{H}}_l^{(m)}\|_F^2}{\|\mathbf{H}_l^{(m)}\|_F^2}, \quad (16)$$

with the networks being optimized using Adam optimization, with a batch size of 64 and a total of 200 training epochs.

4) *Online Deployment*: After the offline training, the trained CNN and UNet are deployed at the receiver for online channel estimation. In this phase, the networks trained for a single subcarrier are applied for each subcarrier $n \in \{1, \dots, N^{(m)}\}$. As the output, the CNN and UNet provide the estimated channel matrix $\bar{\mathbf{H}}^{(m)}[n]$, generated through the mappings given by (14) and (15), respectively. The networks are robust and trained for different wireless channel conditions, in terms of K -factor and SNR. Therefore, there is no need to retrain the networks for each specific K -factor or SNR scenarios, which saves computing power and storage space. However, both the CNN and UNet are limited with respect to the choice of MIMO antenna configuration. If the network should be applied on a different MIMO configuration, retraining with different $M_{\text{Rx}}^{(m)} \times M_{\text{Tx}}^{(m)}$ parameters is necessary.

IV. SIMULATION-BASED COMPARISON

To assess the performance of the proposed channel estimation methods, we simulate the normalized channel estimation and interpolation MSE, and the achievable SE in a frequency-selective channel. The simulation parameters are listed in Tab. I, while the channel is modeled using the 3GPP urban macro for the LOS scenario with an adjustable K -factor, as described in [15]. Based on the measurement-validated similarity of K -factors reported in [18], the scaling factor c_K is set to 1, defining the mmWave K -factor as $K^{(m)} = K^{(s)}$.

In addition to the MRC method discussed in Section III-A and the ML methods presented in Section III-B, we show the performance of methods that rely solely on in-band mmWave information. Specifically, we include a non-ML baseline, where only the mmWave channel estimate $\bar{\mathbf{H}}^{(m)}[n] = \tilde{\mathbf{H}}^{(m)}[n]$

TABLE I
SIMULATION PARAMETERS

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
MIMO Configuration	8×8	8×8
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
Cyclic Prefix t_{CP}	1.19 μs	0.59 μs

is utilized. Furthermore, we show the performance of in-band CNN and in-band UNet methods, which exclude both out-of-band estimate $\hat{\mathbf{H}}^{(m)}[n]$ and the K -factor $\tilde{\mathbf{K}}^{(s)}$. For these methods, we consider $\bar{\mathbf{H}}^{(m)}[n] = f_{\Phi_C}(\tilde{\mathbf{H}}^{(m)}[n], \mathbf{\Gamma}^{(m)}; \Phi_C)$ for the in-band CNN and $\bar{\mathbf{H}}^{(m)}[n] = f_{\Phi_U}(\tilde{\mathbf{H}}^{(m)}[n], \mathbf{\Gamma}^{(m)}; \Phi_U)$ for the in-band UNet. Additionally, we show the performance for the case of perfect channel state information (CSI) ($\bar{\mathbf{H}}^{(m)}[n] = \mathbf{H}^{(m)}[n]$) to establish an upper performance bound.

The normalized channel estimation and interpolation MSE is represented by

$$\text{NMSE} = \frac{1}{L_r N^{(m)}} \sum_{l=1}^{L_r} \sum_{n=1}^{N^{(m)}} \frac{\|\mathbf{H}_l^{(m)}[n] - \bar{\mathbf{H}}_l^{(m)}[n]\|_F^2}{\|\mathbf{H}_l^{(m)}[n]\|_F^2}, \quad (17)$$

where L_r denotes the number of realizations and l indicates a specific channel realization. The achievable SE, averaged over $N^{(m)}$ subcarriers and expressed in bits/s/Hz, is defined by

$$\text{SE} = \frac{1}{N^{(m)}} \sum_{n=1}^{N^{(m)}} \sum_{\mu=1}^{\ell_{\max}} \log_2(1 + \text{SINR}_{\mu}[n]), \quad (18)$$

where the effective signal-to-interference-and-noise ratio (SINR) for the stream μ is given by [13]

$$\text{SINR}_{\mu}[n] = \frac{|\bar{\mathbf{G}}_{\mu,\mu}^{(m)}[n]|^2}{\sum_{\substack{\nu=1 \\ \nu \neq \mu}}^{\ell_{\max}} |\bar{\mathbf{G}}_{\mu,\nu}^{(m)}[n]|^2 + (\sigma_{\mu}^2[n] + \sigma_{w^{(m)}}^2) \|\bar{\mathbf{Q}}_{:, \mu}^{(m)}[n]\|^2}. \quad (19)$$

In (19), $\bar{\mathbf{G}}_{\mu,\nu}^{(m)}[n]$ with $\mu, \nu \in \{1, \dots, \ell_{\max}\}$ represents the elements of the channel gain matrix $\bar{\mathbf{G}}^{(m)}[n] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ for the n -th subcarrier, defined as

$$\bar{\mathbf{G}}^{(m)}[n] = \left(\bar{\mathbf{Q}}^{(m)}[n]\right)^H \mathbf{H}^{(m)}[n] \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n]\right)^{1/2}. \quad (20)$$

Moreover, $\sigma_{\mu}^2[n]$ denotes the diagonal entries (variance) of the estimation error covariance $\bar{\mathbf{C}}_{\varepsilon}^{(m)}[n] = \frac{1}{\ell_{\max}} \bar{\varepsilon}^{(m)}[n] \left(\bar{\varepsilon}^{(m)}[n]\right)^H$ for the n -th subcarrier. The error $\bar{\varepsilon}^{(m)}[n]$ quantifies the difference between the channel gain matrix computed with the estimated precoder/combiner

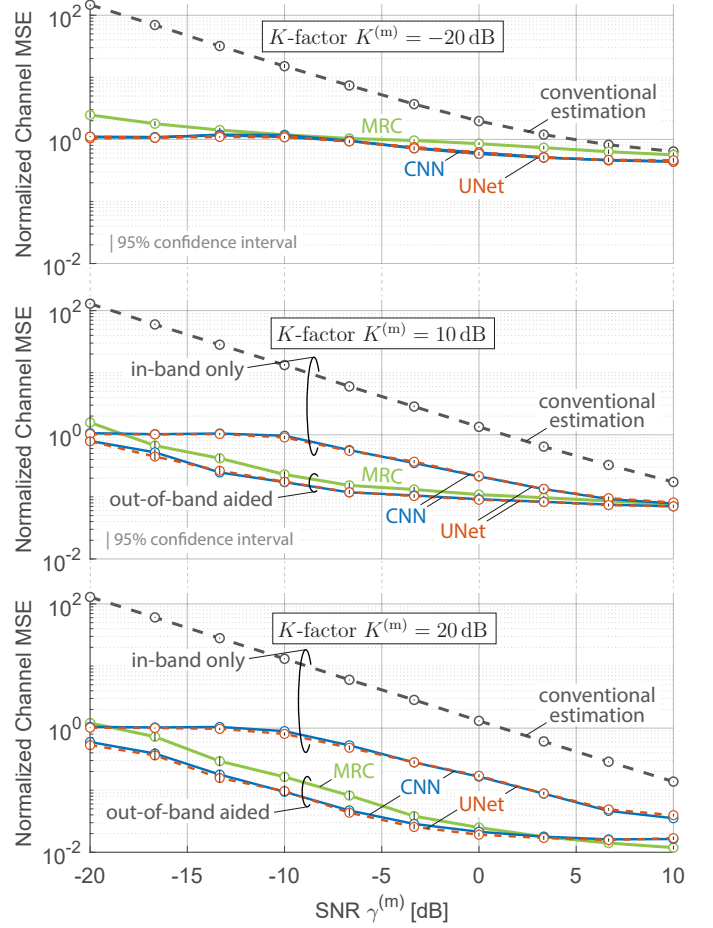


Fig. 4. The proposed out-of-band aided CNN and UNet channel estimation methods achieve the lowest channel MSE across all K -factors. The small vertical bars within the circular markers indicate the 95% confidence intervals.

$\bar{\mathbf{G}}^{(m)}[n]$, and the channel gain matrix obtained using the ideal precoder/combiner $\mathbf{G}^{(m)}[n]$.

In the first simulation, we analyze performance as a function of SNR, with results shown in Fig. 4 for $L_r = 100$ realizations in terms of normalized channel MSE. The normalized channel MSE decreases with increasing SNR, regardless of the estimation method or K -factor. Moreover, all estimation methods consistently outperform the conventional non-ML approach across the entire SNR range and for all K -factors.

Results at K -factor of -20 dB: In this scenario, the in-band and out-of-band aided ML methods exhibit comparable performance, all outperforming the MRC method across the entire SNR range.

Results at K -factor of 10 dB: The MRC method outperforms the in-band CNN and UNet methods, highlighting the significance of missing out-of-band information. However, out-of-band aided CNN and UNet methods maintain comparable performance, outperforming the MRC at all SNR levels.

Results at K -factor of 20 dB: While the MRC method outperforms the in-band CNN and UNet methods, it underperforms relative to the out-of-band-aided CNN and UNet approaches. The out-of-band-aided ML methods demonstrate

TABLE II
DATASET PARAMETERS

Parameter	Value
Training Samples L_{train}	500 000
Test Samples L_{test}	500 000
SNR $\gamma^{(m)}$ [dB]	$\mathcal{U} \sim [-20, 10]$
K -factor $K^{(m)}$ [dB]	$\mathcal{U} \sim [-20, 30]$
AoD ϑ , AoA φ [deg]	$\mathcal{U} \sim [-90, 90]$

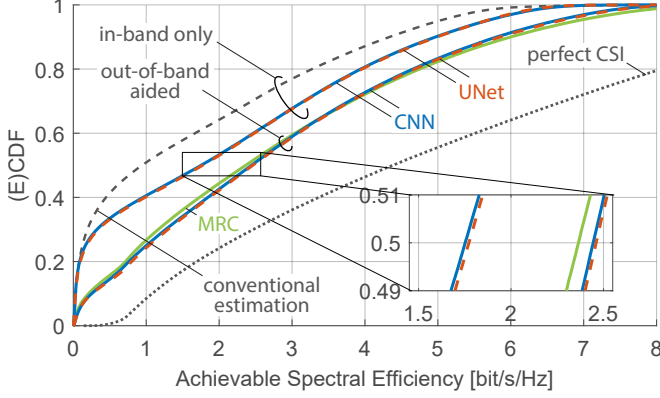


Fig. 5. The proposed out-of-band aided CNN and UNet channel estimation methods achieve a 3.35% and 4.03% higher median SE, respectively, compared to the MRC method.

comparable performance across all SNR levels.

In the next simulation, we generate a dataset with L_{train} and L_{test} channel realizations. For each realization, the SNR $\gamma^{(m)}$, K -factor $K^{(m)}$, AoD ϑ and AoA φ are mutually independent and uniformly distributed within the ranges specified in Tab. II. The cumulative distribution function (CDF) of the achievable SE for all L_{test} test samples is shown in Fig. 5. For in-band methods, the CNN and UNet architectures achieve 86% and 88% higher median SE, respectively, than the non-ML conventional method. However, substantial gains in achievable SE are observed with out-of-band aided channel estimation. The MRC achieves 151% higher median SE, increasing to 159% with the out-of-band aided CNN and 161% with the out-of-band aided UNet. Overall, in both in-band only and out-of-band aided cases, the UNet slightly outperforms the CNN architectures in terms of median SE. However, this gain in SE comes at the cost of increased computational complexity due to the more complex design of the UNet.

V. CONCLUSION

In this paper, we propose two novel deep-learning channel estimation methods for mmWave MIMO systems that leverage sub-6 GHz out-of-band information. The proposed CNN and UNet architectures, when combined with out-of-band information, increase median achievable SE by approximately 3-4% compared to the non-ML MRC method. While both architectures exhibit comparable performance on a large scale, the UNet architecture demonstrates a slight advantage over the CNN, due to its ability to capture fine-grained details.

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